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Broadband low-frequency ventilated acoustic metamaterial using decorated membrane-integrated Helmholtz resonator

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ABSTRACT

Ventilated acoustic metamaterials are critical for balancing noise control and airflow in engineering systems, but traditional designs are plagued by narrow bandwidth and bulky structures when targeting low-frequency noise. Herein, we propose a compact ventilated acoustic metamaterial unit that integrates a Helmholtz resonator (HR) with a decorated membrane resonator (denoted as HRDMR hereinafter). This structure introduces a novel coupling mechanism between a HR and a mass-loaded, tension-insensitive membrane, leveraging the synergistic resonance between cavity air and decorated membrane vibration to achieve broadband low-frequency sound insulation while maintaining unobstructed airflow. Finite element simulations and transmission loss tube experiments demonstrate that a 5-unit cascaded HRDMR array features an ultra-compact size exhibiting sub-wavelength dimensions ($\lambda/16$ at 500 Hz) and achieves a sound transmission loss exceeding 10 dB over the frequency range of 489–1058 Hz. Compared to traditional HRs, this design achieves an approximately 20% bandwidth expansion. Furthermore, this design exhibits robust parameter stability, retaining performance across its operating band even when the Young modulus of the membrane varies within 14–20 MPa (a 30% variation). This work provides a high-performance and ultra-compact solution for low-frequency noise insulation in ventilation ducts, vehicle cabins, and other airflow-dependent engineering applications.

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I. INTRODUCTION

Rapid industrialization has established noise pollution as a significant and widespread problem.[1–3] Low-frequency noise (<1000 Hz), characterized by its long wavelength and strong penetrability, is particularly detrimental to human health and challenging to attenuate using conventional methods.[4] Against the backdrop of rapid economic development, the automotive industry—particularly the new energy vehicle sector—faces growing consumer demands for enhanced cabin comfort, which intensifies the need for effective noise control technologies. In the field of vehicle noise, vibration, and harshness (NVH) control, human perception of noise reduction follows a clear pattern: a reduction of 3–5 dB represents the minimum perceptible improvement, while a reduction of ≥8 dB significantly enhances the sense of quietness and effectively mitigates hearing fatigue during prolonged driving.[5–7] Traditional sound insulation materials, when used to attenuate low-frequency noise, typically require bulky structures commensurate with the acoustic wavelength; these often impede airflow and conflict with practical engineering constraints. Moreover, the implementation of noise reduction devices is often constrained by strict spatial limitations, necessitating compact designs. In applications such as air conditioning systems, engine compartments, and various motor systems, effective heat dissipation and air exchange are crucial for operational stability,[8] while noise control directly impacts user comfort. Consequently, achieving a balance between ventilation efficiency and noise control represents a significant challenge in the design of industrial ventilation systems.[9]

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The emergence of acoustic metamaterials (AMs) offers an innovative solution to this challenge through sub-wavelength structural designs that manipulate sound waves. Unlike traditional materials, which rely primarily on mass or porosity, AMs typically composed of periodic or aperiodic units[10–13] and are engineered to utilize local resonance and tailored dispersion to block, absorb, or redirect noise at specific target frequencies.[14–17] Furthermore, the rapid advancement of additive manufacturing technology has significantly facilitated the design and fabrication of AMs.[18] Established AM designs, including Helmholtz resonators (HRs),[19–22] Fabry–Perot (FP) tubes,[23–26] micro-perforated plates (MPPs),[27–30] and membrane resonators,[16,31–34] achieve low-frequency sound attenuation primarily via strong resonance effects. However, these resonant designs are often constrained by a narrow bandwidth, which limits their practical utility in scenarios requiring broad-spectrum noise control.

Recent research on AMs has sought to overcome these limitations through multi-resonant unit integration and hierarchical designs, which broaden the bandwidth while maintaining compactness. For instance, Li *et al.*[35] developed a double-layer membrane AM that incorporated Helmholtz resonators. By adjusting the cavity height and the positioning of the membrane defect, this design achieved a combination of lightweight, load-bearing, and sound-insulating properties suited to engineering applications. Similarly, Wang and Bennett[36] reported a broadband acoustic metasurface that combined MPPs with extended-neck Helmholtz resonators, achieving high-efficiency noise absorption below 200 Hz and broadband absorption from 200 to 1000 Hz. Choi and Jeon[37] proposed a sound-absorbing metasurface comprising a membrane and HRs, which enabled perfect low-frequency sound absorption through hybrid resonance. They established a theoretical framework and optimized the membrane's position within the structure, achieving nearly perfect absorption at a target frequency of 500 Hz. Advancements in ventilated metamaterials are also notable Meng *et al.*[38] achieved high-efficiency low-frequency sound absorption through optimized structural design. Nguyen *et al.*[21] proposed a ventilated acoustic metamaterial that employs slit Helmholtz resonators (SHRs) to achieve efficient sound insulation within the 0.48–0.95 kHz range; however, the single-resonance mechanism limits its bandwidth, which remains to be broadened. Fusaro *et al.*[39] designed and tested various configurations of acoustic metamaterials. Through integrated numerical simulations and experimental analyses, they investigated the impact of airflow on acoustic performance, thereby providing both valuable theoretical and experimental support for ventilated metamaterials. Despite these advancements, challenges remain in achieving structural compactness, bandwidth scalability, and manufacturing complexity.

To address these, this work introduces a ventilated acoustic metamaterial unit comprising an HR coupled with a mass-loaded, tension-insensitive decorated membrane, hereafter termed HRDMR. Operating at deep sub-wavelength scales along the direction of wave propagation, it achieves a compact, low-cost, and mechanically simple structure. In comparison with the traditional Helmholtz resonator or pure membrane absorber, the proposed structure provides a broader bandwidth. Moreover, HRDMR exhibits higher ventilation efficiency than conventional ventilation metamaterials,[40,41] owing to its installation on the sidewall of the pipeline. A detailed performance comparison with some reference structures is provided in Table VI in the Appendix. This table lists the size, effective bandwidth, and ventilation efficiency of some other studies from references. Through parametric optimization and an iterative design process for multiple units, we developed a cascaded array of five HRDMRs, which demonstrated a significant sound transmission loss (STL) exceeding 10 dB over a frequency range of 489–1058 Hz. Owing to its modular design and ventilation compatibility, the HRDMR is suited for applications such as engine compartments and heating, ventilation, and air conditioning ducts, where stable airflow and noise suppression are essential.

II. RESULTS AND DISCUSSION

A. Design of the HRDMR unit

In the low-frequency range, conventional HRs exhibit only a single sound insulation peak with a relatively narrow bandwidth. To overcome this limitation, a mass-loaded decorated membrane is incorporated. Membrane-type acoustic metamaterials typically exhibit multiple resonant modes, which can be leveraged to broaden the sound insulation bandwidth. The incorporation of such a mass-loaded decorated membrane into a Helmholtz resonator induces multiple sound insulation peaks, thereby significantly broadening the bandwidth of a single unit. Furthermore, traditional sound insulation materials often struggle to balance acoustic performance with ventilation efficiency. Consequently, the proposed metamaterial unit—hereafter termed a Helmholtz resonator with a mass-loaded, tension-insensitive decorated membrane (HRDMR)—is engineered to integrate sound insulation with ventilation capabilities, enabling simultaneous noise suppression and unobstructed airflow in pipelines. As illustrated in Fig. 1(a), the HRDMR unit is mounted on the pipe wall, allowing incident noise and airflow to propagate axially while preserving ventilation efficiency. Sound attenuation is achieved through acoustic interactions within the unit's internal structure. Detailed geometry is presented in the cross-sectional view [Fig. 1(b)], which identifies the key geometric parameters characterizing the HRDMR. The unit's shell is fabricated from polylactic acid (PLA), the decorated membrane is composed of thermoplastic polyurethane (TPU), and the added masses are made of aluminum. The detailed geometric parameters are listed in Table I, and the physical properties of each component are provided in Table II.

The HRDMR comprises a rigid shell with a uniform wall thickness t and an integral unit thickness w_c. The neck region, represented by the translucent block on the left in Fig. 1(b), features a channel of width l_n that connects the main airflow path to the resonator cavity. Similarly, the right translucent block represents the resonator cavity, which is partitioned by the decorated membrane into two chambers: an open cavity (height h_c) adjacent to the neck and a closed cavity (height h_s) on the opposite side of the membrane. The dynamic interaction between the open cavity and closed cavity modulates the acoustic response through the membrane's vibration. The TPU membrane has a thickness t_m, with two symmetrically distributed semicircular mass blocks attached to its surface. Each mass block has a radius r_a and thickness t_a, and the distance between their centers is x_m. Unlike existing hybrid structures (e.g., the double-layer membrane with HRs[35] or

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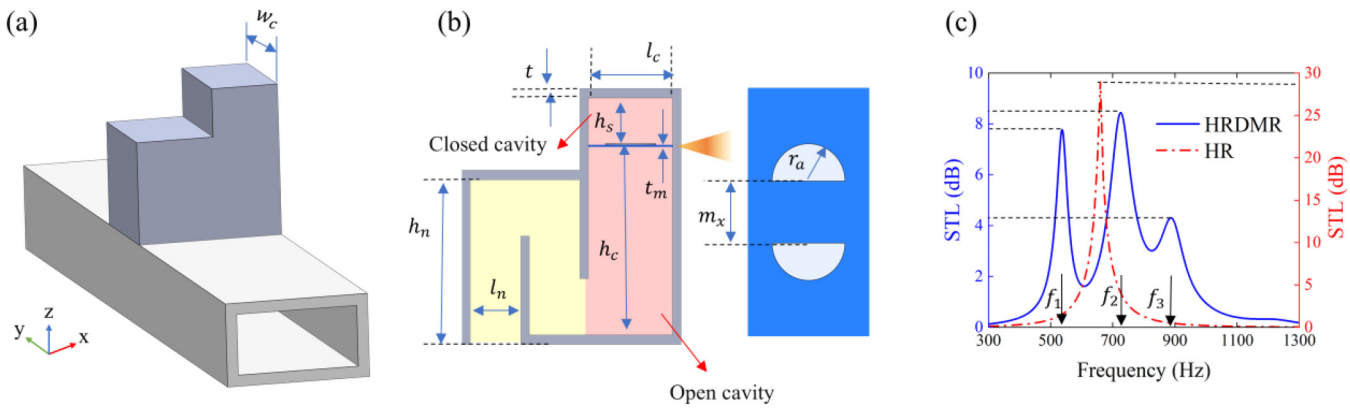

FIG. 1. Design and acoustic properties of the HRDMR unit. (a) Schematic of the HRDMR unit structure, with the bottom rectangle representing the ventilation channel. (b) Cross-sectional view defining the key geometric parameters of the HRDMR unit. (c) STL curve of an HRDMR unit and an HR unit, where the frequencies corresponding to f_1, f_2, and f_3 are the three resonance frequencies of the HRDMR unit. The solid line is the HRDMR correlation curve (left Y axis), and the dotted line is the HR correlation curve (right Y axis).

membrane-integrated HRs[42]) that often require precise tension control of membranes or complex assembly of discrete components, the HRDMR employs a tension-insensitive design and a modular architecture: specifically, the TPU membrane is securely bonded without requiring tension adjustment. This design simplifies manufacturing, reduces the requisite assembly precision, and enhances potential for large-scale industrial production.

STL is a critical metric in engineering for evaluating the performance of sound insulation materials, as it quantifies the reduction in sound energy transmitted through a structure. It is defined as the logarithmic ratio of the incident sound energy to the transmitted sound energy, expressed as

$$\mathrm{STL} = 10\log_{10}\left(\frac{W_i}{W_t}\right). \tag{1}$$

Here, W_i is the incident sound power and W_t is the transmitted sound power.

TABLE I. Detailed geometric parameters of HRDMR.

Parameter description	Symbol	Unit (mm)
Neck height	h_n	30
Neck passage width	l_n	12
Thickness of membrane	t_m	0.4
Open cavity height	h_c	40
Close cavity height	h_s	10
Cavity width	l_c	20
Unit depth	w_c	40
Mass thickness	t_a	0.3
Mass radius	r_a	6
Mass block spacing	x_m	10
Wall thickness	t	2

The frequency-dependent STL spectra for a single HRDMR unit and a single HR unit are compared in Fig. 1(c). It should be noted that the HR mentioned here refers to the cavity after the decorated membrane is removed from the HRDMR. As shown, the HR unit exhibits a single sound insulation peak at approximately 660 Hz, with an effective bandwidth (STL > 10 dB) of about 50 Hz. Although the STL peak value of HRDMR is lower, its three sound insulation peaks span a broader frequency range (536–890 Hz), resulting in a wider effective bandwidth. This broader bandwidth can compensate for the lower peak value through superposition. However, the STL peak value for the HR is higher than that of the HRDMR unit. From an energy perspective, excessively high STL values offer limited practical significance, and the broader attenuation band is often more valuable. It is evident from the figure that the HRDMR unit exhibits three distinct sound insulation peaks below 1000 Hz, at 536, 726, and 890 Hz, respectively. The three peaks of the HRDMR correspond to specific modes: (1) symmetric vibration of the mass-loaded membrane (536 Hz), (2) localized vibration of the membrane between mass blocks (726 Hz), and (3) air resonance in the open cavity (890 Hz), as validated by the modal analysis in Sec. II B.

B. HRDMR unit analyses

To comprehensively characterize the acoustic performance of the HRDMR, full-wave acoustic simulations were performed using the pressure acoustics module in COMSOL Multiphysics. The inlet

TABLE II. Physical properties of each component.

Component name (material)	Density (kg/m^3)	Young's modulus (MPa)	Poisson's ratio
Rigid housing (PLA)	1 430	2 000	0.4
Mass (aluminum)	2 700	70 000	0.33
Membrane (TPU)	1 200	20	0.43

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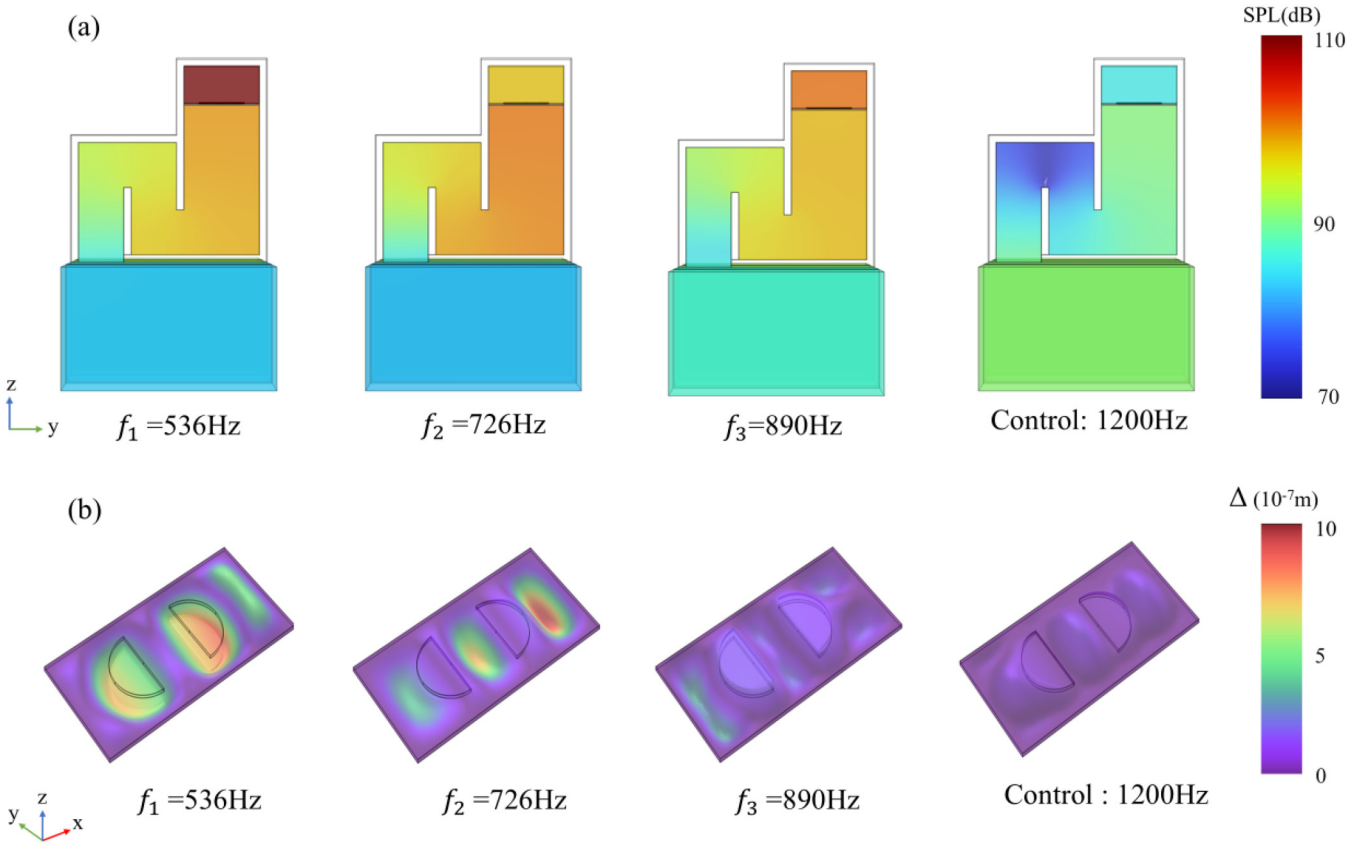

FIG. 2. Simulated sound response of the HRDMR unit. (a) Sound pressure level distribution at 536, 726, 890, and 1200 Hz. (b) Membrane displacement distribution at the same frequencies. It should be noted that 1200 Hz is a non-resonant frequency, while 536, 726, and 890 Hz correspond to three resonance peaks shown in Fig. 1(c).

boundary was excited by an incident plane wave at normal incidence, and the outlet boundary was terminated with a perfectly matched layer (PML) to prevent spurious reflections. All structural boundaries (PLA shell, membrane, mass blocks) were set to rigid except for the TPU membrane, which was modeled as a flexible structure with linear elastic properties. To improve computational efficiency, narrow region acoustic modeling was employed in lieu of the thermal-viscous dissipation module. The background medium was assigned as air, with a density of 1.12 kg/m^3 and a sound speed defined as 343 m/s.

Figure 2(a) illustrates the sound pressure level distribution within the HRDMR unit at the resonant frequencies of 536, 726, and 890 Hz, as well as at a non-resonant frequency of 1200 Hz. The corresponding membrane displacement distributions at these frequencies are shown in Fig. 2(b). At 536 Hz, the decorated membrane primarily undergoes symmetric vibration. The motion of the two mass blocks couples with the membrane's oscillation, effectively exciting air vibrations within the closed cavity and thereby enhancing acoustic energy dissipation. At 726 Hz, the two mass blocks act as effective fixed points, restricting membrane vibration in the region beneath the mass blocks. Consequently, the membrane vibrates more intensely between the two mass blocks and near its front and rear edges. At 890 Hz, which corresponds to a lower STL peak of approximately 4.5 dB, membrane vibration is relatively weak, and sound attenuation is primarily attributable to the resonant vibration of the air within the cavity. In contrast, at the non-resonant control frequency of 1200 Hz, both the cavity air and the membrane exhibit minimal vibration, leading to an STL value near 0 dB. These modes directly correspond to the STL peaks in Fig. 1(c), validating the coupling mechanism between membrane and cavity dynamics.

Notably, when an incident sound wave excites the HRDMR's resonance mode, high sound pressure accumulates within the entire enclosed cavity, indicating that both the decorated membrane and cavity air participate in the acoustic response. In contrast to traditional HRs with rigid boundaries, which produce a single STL peak via neck–cavity resonance, the HRDMR utilizes a decorated membrane to introduce additional energy dissipation mechanisms. On one hand, the decorated membrane's intrinsic resonance contributes to additional sound energy dissipation. On the other hand, sound waves in the open cavity are into the closed cavity via the membrane, thereby enhancing sound energy attenuation within the structure. These two mechanisms work synergistically to enhance

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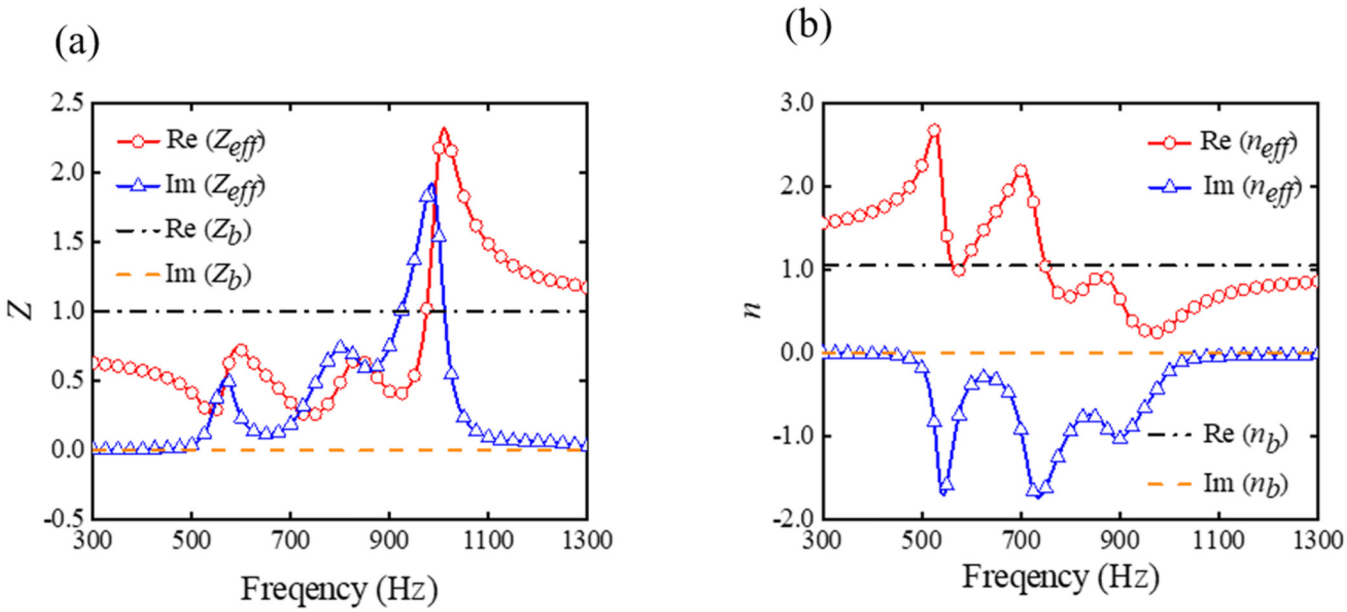

FIG. 3. Retrieved effective acoustic parameters: (a) relative refractive index n and (b) relative acoustic impedance Z.

the HRDMR's broadband and tunable sound insulation performance. Notably, due to its specialized design, the HRDMR has a thickness of only $\lambda/16$ in the wave propagation direction (x axis) at 500 Hz. This compact form factor enables deployment in space-constrained environments, significantly expanding its potential engineering applications.

Given the significant impedance mismatch between air and the PLA outer shell, the shell surface was modeled as a perfectly rigid boundary. The transmission (T) and reflection (R) coefficients of the structure were calculated using the four-microphone method,[43] which is based on the principle of acoustic impedance tubes. Subsequently, the effective acoustic parameters were retrieved using the inversion method.[44] These effective properties, derived from the simulated T and R, provide insights into the wave manipulation mechanisms of metamaterials. The effective relative acoustic impedance Z is given by[43]

$$Z_{\text{eff}} = \frac{\chi}{1 - 2R + R^2 - T^2}. \tag{2}$$

The parameter χ is defined as follows:

$$\chi = \pm\sqrt{(R^2 - T^2 - 1) - 4T^2}. \tag{3}$$

Here, we choose the root that provides a positive solution in Eq. (2).[43,45] Furthermore, the acoustic refractive index is expressed as

$$n_{\text{eff}} = \frac{-j\ln\zeta + 2\pi m}{k_i H}, \tag{4}$$

where m is the number of tubes, and a minimum thickness of the AM that corresponds to $m = 0$ is used during the retrieval of effective parameters. ζ is defined as

$$\zeta = \frac{1 - R^2 - T^2 + \chi}{2T}. \tag{5}$$

The effective acoustic properties retrieved via the inversion method are presented in Figs. 3(a) and 3(b). In these figures, the real and imaginary parts for air are shown as dotted and dashed lines, respectively [Re ($Z_{\text{eff}}/\rho c$) = 1, Im ($Z_{\text{eff}}/\rho c$) = 0; Re (n_{eff}) = 1, Im (n_{eff}) = 0]. The corresponding real and imaginary parts for the HRDMR are denoted by solid lines with circular and triangular markers. Figures 3(a) and 3(b) show the effective impedance and effective refractive index profiles near the three sound insulation peaks. At these resonant frequencies, the relative acoustic impedance Re ($Z_{\text{eff}}/\rho c$) fluctuates between 0.8 and 2.2, while peaks of Im ($Z_{\text{eff}}/\rho c$) are 0.5–1.9. This impedance fluctuation range is consistent with the typical range of locally resonant acoustic metamaterials reported by Fokin *et al.*,[43] confirming the reliability of our parameter retrieval. In addition, the effective refractive index Im (n_{eff}) reaches peaks of ~1.8 (536 Hz), ~1.5 (726 Hz), and ~1.0 (890 Hz) at the three resonance frequencies.

Apparently, the equivalent parameter characteristics of HRDMR exhibit significant deviations near the resonance frequencies. For acoustic metamaterials, the magnitude of Im (n_{eff}) directly reflects the rate of acoustic energy attenuation, while $Z_{\text{eff}}/\rho c$ deviating 1 indicates impedance mismatch. Near the three resonances, the effective acoustic impedance of the HRDMR exhibits marked fluctuations, and there are varying degrees of impedance mismatch,

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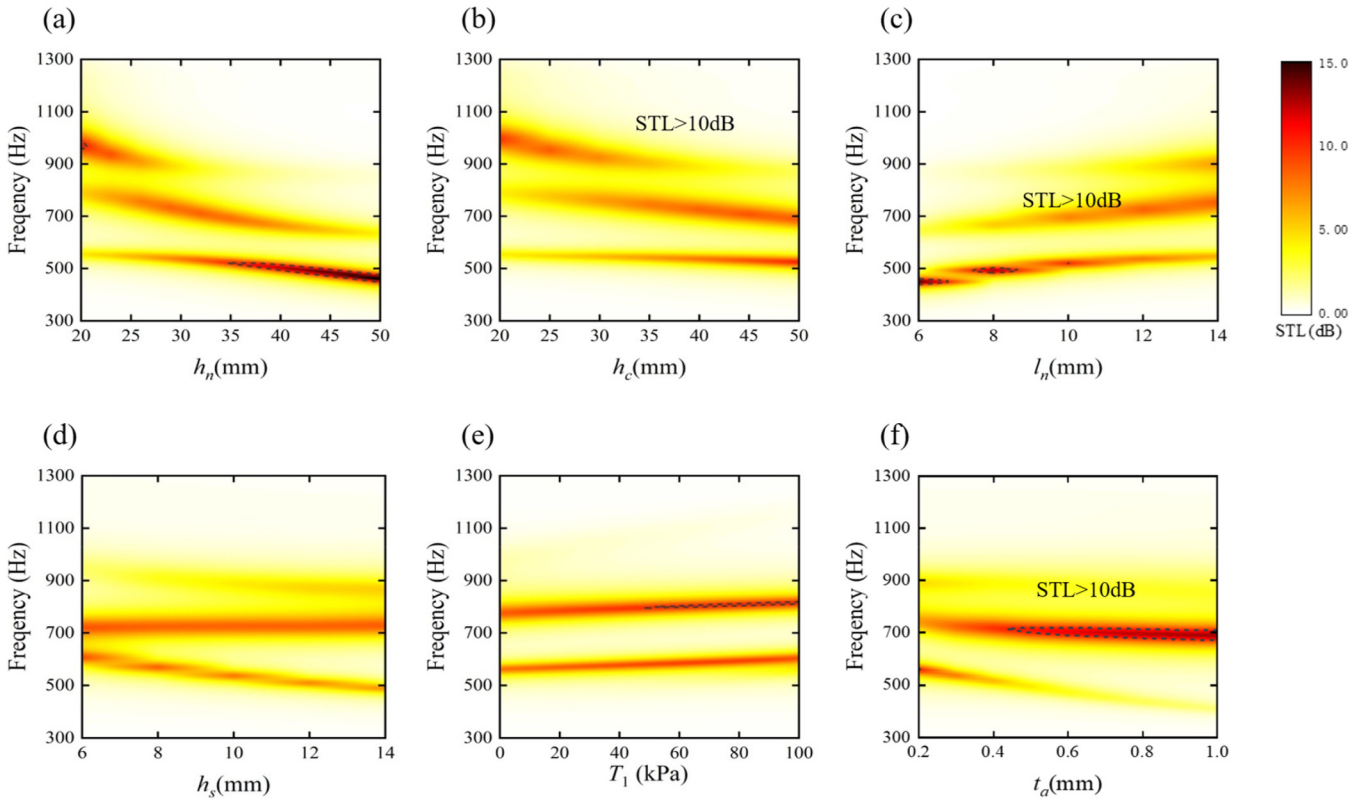

FIG. 4. Simulated STL with key geometric parameters: (a) neck height h_n, (b) open cavity height h_c, (c) neck opening width l_n, (d) closed cavity height h_s, (e) membrane prestress, (f) mass thickness t_a. Other parameters fixed at values in Table I. The dot and line areas indicate STL > 10 dB.

leading to interface reflection. This resonance-induced impedance mismatch underpins deep sub-wavelength resonance and enables broadband sound insulation.

C. Influence of geometric parameters

The acoustic performance of the HRDMR is highly dependent on its geometric parameters. A systematic investigation was conducted to evaluate the influence of key geometric parameters on the HRDMR's STL. By tuning these parameters, the acoustic characteristics of the unit can be tailored to a desired target frequency range.

As shown in Fig. 4(a), increasing h_n shifts the STL peaks toward lower frequencies. This is because increasing h_n lengthens the neck channel, which increases the acoustic mass and, thus, lowers the resonance frequency. Figure 4(b) demonstrates that an increase in h_c enlarges the cavity volume. This enhances the air vibration amplitude within the cavity, resulting in greater energy dissipation and a shift of the resonance frequencies toward lower values. In contrast, Fig. 4(c) shows that increasing l_n elevates all three resonance frequencies and significantly broadens the bandwidth of the STL peak corresponding to f_2. This effect is attributed to the wider neck channel, which reduces acoustic mass (increasing resonance frequency) and lowers flow resistance, thereby facilitating broader coupling. Figure 4(d) shows that increasing h_s reduces the frequencies corresponding to f_1 and f_3, while the value of f_2 increases slightly. This indicates that the dissipation of acoustic energy in the closed cavity has a minimal contribution to the STL peak f_2, which is consistent with the previous analysis.

During the fabrication and installation process, a certain degree of prestress inevitably develops on the membrane surface. Therefore, it is necessary to systematically study this effect through simulation. Figure 4(e) illustrates the influence of membrane prestress, ranging from 0 to 100 kPa, on the unit's sound insulation performance. As shown, the application of membrane prestress has a minimal impact on the sound insulation performance of the unit. Additionally, a prestress of 100 kPa represents a relatively large value for the membrane, further indicating that the element is insensitive to tension. Due to the mass-loading effect, increasing the mass reduces the resonance frequency of the decorated membrane—a trend also observed in the HRDMR. As t_a increases, the mass of the mass block increases, enhancing the system's inertia. For a given stiffness, the increased mass is less prone to vibration, thereby lowering the system's resonance frequency. As shown in Fig. 4(f), increasing t_a shifts all three resonance peaks to lower frequencies, with f_1 showing a more pronounced shift. This indicates

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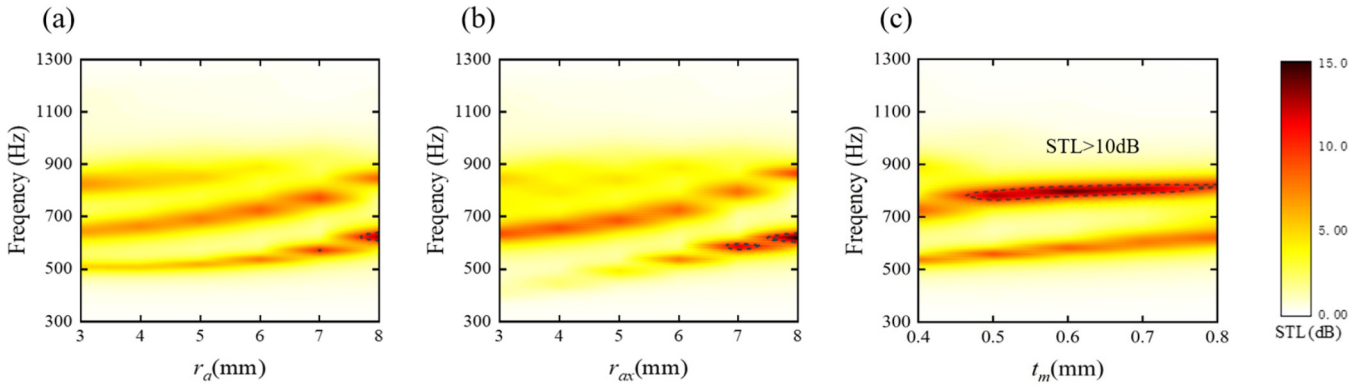

FIG. 5. Simulated STL with decorated membrane geometric parameters: (a) mass radius r_a, (b) redefined radius of mass block r_{ax} (the mass of the mass block remains unchanged when r_{ax} changes), (c) thickness of membrane t_m. Other parameters fixed at values in Table I. The dot and line area indicate STL > 10 dB.

that the STL peak f_1 is primarily dominated by membrane resonance, while membrane vibration contributes little to STL peaks f_2 and f_3, consistent with our initial hypothesis.

Figure 5(a) illustrates the influence of the mass block radius, r_a, on the STL performance. As r_a increases, all three sound insulation peaks shift to higher frequencies. However, this trend contrasts with that shown in Fig. 4(f), indicating that an increase in r_a elevates the HRDMR's peak frequencies despite the concurrent increase in mass. To decouple these effects, a new parameter r_{ax} was defined, representing the radius of a mass block with a constant mass (achieved by adjusting its thickness proportionally). The influence of the redefined r_{ax} on STL performance is shown in Fig. 5(b), confirming our hypothesis. The increased radius enhances the contact area between the mass blocks and the decorated membrane, strengthening their interaction and increasing the system's stiffness. For a constant mass, increased stiffness raises the resonance frequency. Figure 5(c) shows the effect of membrane thickness variations on STL performance. Increasing t_m shifts the STL peak frequencies to higher values. This shift arises from the combined effects of increased mass and stiffness of the membrane. Similar to the mass blocks, increased membrane mass tends to lower the STL frequency, while increased stiffness raises it. Therefore, the stiffness effect dominates the observed frequency shift in Fig. 5(c).

In order to understand the influence of a certain parameter on the peak-to-peak frequency of sound insulation more intuitively, the sensitivity coefficient S_i is introduced for analysis and comparison. The calculation formula of S_i is as follows:

$$S_i = \frac{\Delta f_i / f_{oi}}{\Delta x / x} \times 100 \quad (i = 1, 2, 3). \tag{6}$$

Here, $i = 1, 2, 3$ correspond to the three resonance frequencies mentioned above. The term Δf_i represents the change in the frequency of the sound insulation peak, and f_{oi} represents the initial frequency of the sound insulation peak, where x denotes a specific geometric size parameter, such as h_s, h_n, and l_n. The calculated sensitivity coefficients are summarized in Table III. A larger magnitude of the sensitivity coefficient indicates that the corresponding resonance frequency is more significantly influenced by variations in that geometric parameter. Furthermore, the sign of the coefficient indicates the nature of the correlation: a positive value signifies that the resonance frequency increases with the parameter (direct proportionality), while a negative value signifies a decrease (inverse proportionality). Such as increasing h_n lengthens the neck channel, thereby lowering the unit's sound insulation frequency. Conversely, increasing l_n widens the neck channel, raising the unit's sound insulation frequency. An increase in t_a adds mass to the mass block, increasing the system's inertia and consequently reducing the sound insulation frequency. The increase in r_{ax} will increase the contact area between the mass and the membrane, boosting the system's stiffness and thus raising the sound insulation frequency.

D. Experimental validation

The resonant frequency of the HRDMR can be tuned by adjusting its geometric parameters to achieve effective sound attenuation within targeted frequency bands. To attain a broader noise reduction bandwidth, five distinct units were designed through parametric analysis and iterative optimization. These units were

TABLE III. Sensitivity coefficient table.

Symbol	S_1	S_2	S_3	Above
h_n	−11.9	−30.4	−18.6	−20.3
h_c	−8.10	−19.3	−19.3	−15.6
l_n	16.7	12.6	3.11	10.8
h_s	−14.8	0.82	−6.11	−6.70
t_a	−6.71	−1.72	−1.13	−3.19
r_a	13.7	19.0	7.01	13.2
r_{ax}	30.7	22.9	9.21	20.9
t_m	14.8	12.3	1.12	9.41

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TABLE IV. Five units with different structural parameters were tested in a straight tube.

No.	1	2	3	4	5
h_s (mm)	10	10	10	10	6
h_n (mm)	45	35	26	18	35

then cascaded to form a sound insulation device with a total length of 210 mm, width of 52 mm, and height of 50 mm. The varying parameters among the five units are listed in Table IV; only h_s and h_n differ, while all other parameters are consistent with those in Table I. This section details the experimental for STL measurement, including the fabrication of the HRDMR prototypes and the description of the experimental setup. The STL measurements were performed using a four-microphone impedance tube system (SZZB-4 acoustic impedance tube, Beijing Institute of Vibration, Beijing, China) to experimentally validate the HRDMR's performance. The four-microphone impedance tube adopts a circular cross section with an inner diameter of 29 mm, which is commonly used in the acoustic test. The tube is divided into three segments: (1) incident segment (length = 280 m); (2) test segment (length = 300 m, for mounting the HRDMR array); and (3) receiving segment (length = 280 m). The sound insulation performance was evaluated using the four-microphone method, which is based on acoustic pressure transfer functions. In this method, four microphones are positioned at specific locations on both sides of the test sample to measure the sound pressure at each point. Utilizing these measurements, we calculate the four characteristics of sound wave transmission and reflection across the component. The sound insulation performance is then quantified using the transmission loss. Photographs of the fabricated samples and the experimental setup are presented in Fig. 6(a).

Figure 6(b) shows the assembled structure of the five-unit HRDMR array, and Fig. 6(c) details the internal structure of a single unit. The components include a rectangular neck, an open cavity, a closed cavity, and a window-type membrane with an attached mass block. The membrane was adhesively bonded between the open and closed cavities, approximating a fixed boundary condition. The units' shell and the connecting piping were fabricated using 3D printing. The sample was securely mounted in the test section using an adapter pipe. A comparison between the experimental and numerical simulation results is presented in Fig. 6(d). In this figure, the slash area indicates where the experimental values exceed the simulation values, while the grid line area indicates where the simulation values exceed the experimental values. It is evident that the HRDMR array achieves STL > 10 dB in the frequency range of 489–1058 Hz, attenuating 90% of incident acoustic energy. As can be seen from Fig. 6(d), the maximum

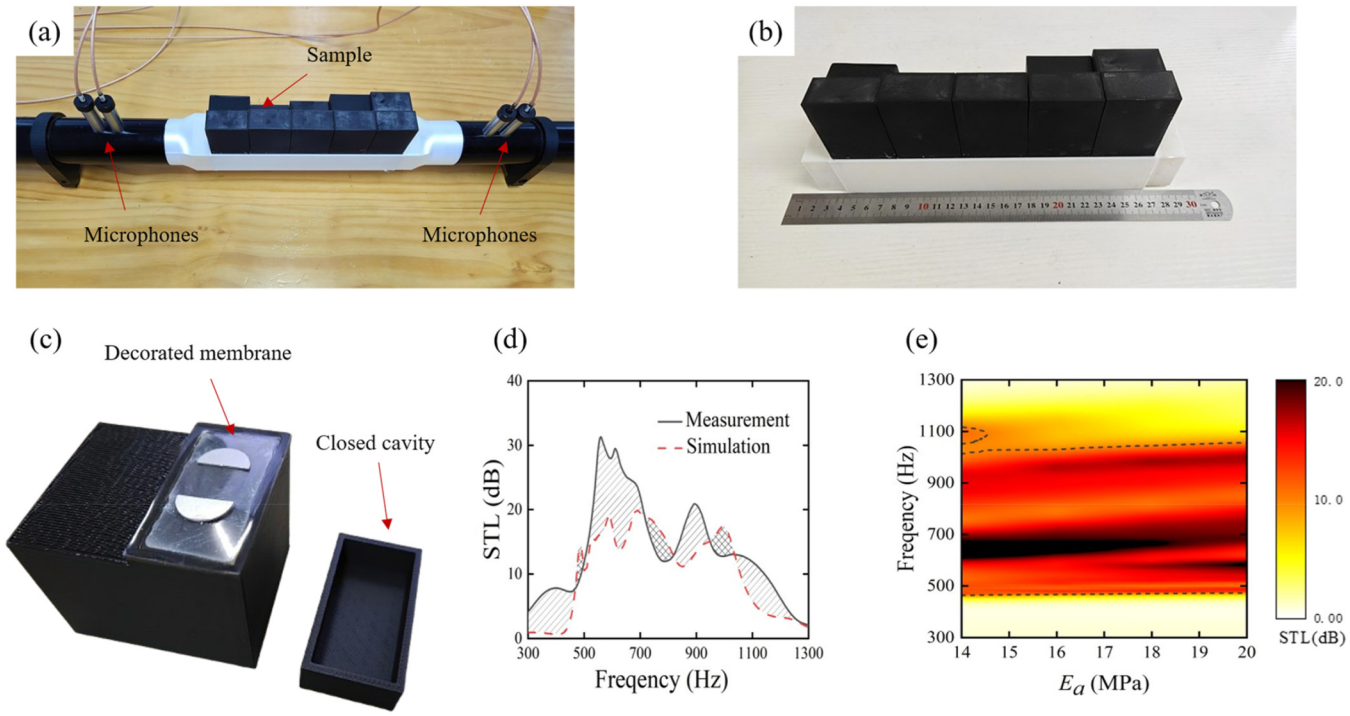

FIG. 6. (a) Photograph of the experiment setup. (b) 3D-printed HRDMR unit assembly sample. (c) The internal structure of one unit. (d) Simulated (dotted line) and measured (solid line) STL of the HRDMR group. (e) HRDMR group simulation STL changes with Young's modulus of the membrane, and other parameters are consistent with the previous discussion.

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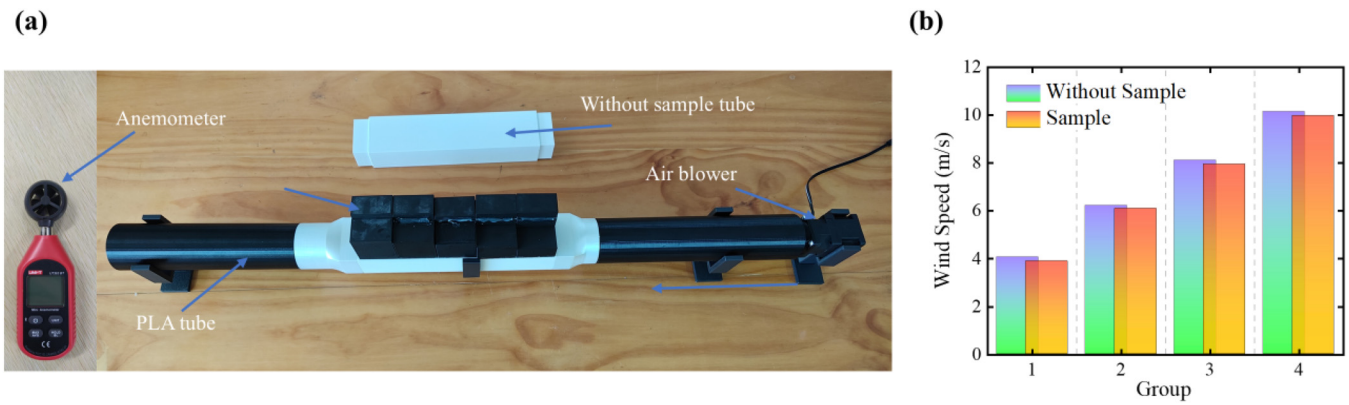

FIG. 7. (a) Schematic diagram of the ventilation measurement experimental device; (b) histograms of air flow rates with and without sample passage. The corresponding target velocities of groups 1, 2, 3, and 4 are 4, 6, 8, and 10 m/s, respectively.

deviation of STL between experiment and simulation is about 15 dB, and the average deviation is 3.5 dB. Although there are some deviations between the simulated and experimental curves, their trends are consistent. The discrepancies between simulation and measurement likely arise from a combination of factors including insufficient 3D printing accuracy, unavoidable air leakage during assembly, uncontrolled thermo-viscous losses in the measurement system, incomplete matching of boundary conditions between experiment and simulation, and non-negligible damping of the device materials.

TPU is a versatile engineering plastic whose physical properties vary with raw material proportions. Consequently, the Young modulus of TPU varies between production batches, and membranes produced via different processes often exhibit property variations that can affect the structure's overall sound insulation performance. Owing to the versatility of COMSOL software, this issue can be examined at the simulation level. As depicted in Fig. 6(e), simulations varying the Young modulus of the TPU membrane reveal that for values between 14 and 20 MPa, the five-unit array maintains STL > 10 dB in the frequency range of 487–1060 Hz, demonstrating good robustness to material property variations. This confirms the robust performance of the HRDMR array across its operating frequency band despite variations in the membrane modulus.

TABLE V. Table of ventilation efficiency.

Group (target wind) speed)	Measured wind speed		Ventilation efficiency (%)
	Sample (m/s)	Without sample (m/s)	
1 (4 m/s)	3.89	4.06	95.8
2 (6 m/s)	6.09	6.21	98.1
3 (8 m/s)	7.94	8.11	97.9
4 (10 m/s)	9.97	10.14	98.3

The ventilation performance of the proposed HRDMR was evaluated using a modified impedance tube setup to compare airflow velocity, as illustrated in Fig. 7(a). The original impedance tube, being a closed system, was replaced with a 3D-printed PLA tube to ensure that the key dimensions of the PLA tube match those of the impedance tube. During the experiment, Vaseline was used to seal the joints between the device components to prevent air leakage. A blower with adjustable speed was positioned at the pipeline inlet to generate a stable and controllable airflow, simulating realistic ventilation conditions. At a distance of 10 mm from the pipeline outlet, an anemometer (UNI-T, Guangdong, China, with a measurement range of 0–30 m/s) with a probe size comparable to the tube's inner diameter was installed to measure the average air velocity over a defined time interval. To ensure measurement reliability, the test was repeated three times under each set wind speed condition, and the average value was used. The airflow test was conducted at four different inlet speeds, ranging from low to relatively high, to assess performance under various operating conditions. The experimental results shown in Fig. 7(b) compare the exit wind speed with and without HRDMR samples. The wind speed attenuation caused by the sample remained below 10%, indicating a minimal impact on airflow. This demonstrates that HRDMR exhibits excellent ventilation performance and low aerodynamic drag, supporting its classification as a ventilated acoustic metamaterial. The specific data corresponding to Fig. 7(b) are provided in Table V, where ventilation efficiency is quantified by the wind speed ratio. In summary, HRDMR achieves a high level of ventilation and sound insulation simultaneously.

Existing acoustic metamaterials generally face the challenge of balancing structural compactness, effective sound insulation bandwidth, and ventilation efficiency (as shown in Table VI in the Appendix). Helmholtz resonance-based metamaterials have either a large volume or limited ventilation efficiency, membrane-type metamaterials are difficult to achieve effective ventilation, and even metamaterials with the membrane-coupled Helmholtz resonance mechanism still need improvements in the compact and sound insulation bandwidth. This work demonstrates superior

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comprehensive performance in terms of compact volume, broadband sound insulation, and high-efficiency ventilation, significantly outperforming existing similar structures. This underscores the innovation and practical application potential of this study in the field of ventilated acoustic metamaterials.

III. CONCLUSIONS

In summary, we have theoretically and experimentally demonstrated a broadband low-frequency ventilated metamaterial. The metamaterial structure comprises Helmholtz resonators (HRs) and a tension-insensitive TPU membrane with additional mass. Both simulation and experimental results have successfully confirmed our theoretical predictions. By adjusting the structural dimensions of the HRDMR unit, the STL peak frequency, sound insulation bandwidth, and average effective STL increment can be precisely controlled. This tunability allows HRDMR units to be tailored to achieve the desired acoustic performance within specific dimensional constraints. Based on this, a noise reduction structure composed of five HRDMR units has been constructed, effectively attenuating over 90% of incident acoustic energy across the broadband range of 489–1058 Hz. It should be noted that this bandwidth was essentially determined at the outset of the structural design. If a lower frequency or wider bandwidth is required, a larger structure would be necessary. This represents a potential direction for future optimization. In the ventilation performance experiment, the structure installed on the sidewall of the pipeline does not cause significant attenuation of gas flow velocity. However, due to experimental limitations, the influence of airflow on the acoustic performance was not investigated. This represents a critical area for future work. It is essential to investigate the impact of low Mach number air swept jets on the performance of acoustic structures in industrial applications. Furthermore, the design at the deep sub-wavelength scale holds significant promise for future ventilated sound insulation applications. Last, the proposed noise reduction structure offers the advantages of lightweight, low cost, and flexible installation, making it well-suited for industrial applications.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jing Nie: Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Fengbin Lin:** Conceptualization (equal); Funding acquisition (lead); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Jinhong He:** Methodology (equal); Project administration (equal); Validation (equal). **Xiaodong Zhang:** Conceptualization (equal); Funding acquisition (lead); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Yang Liu:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal). **Jianxing Yang:** Methodology (equal); Project administration (equal). **Rui Yang:** Investigation (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: COMPARATIVE PERFORMANCE OF VENTILATED INSULATION METAMATERIALS: STL, FREQUENCY, VENTILATION EFFICIENCY, AND VOLUME

Table VI.

TABLE VI. The STL, resonance frequency, ventilation efficiency, and volume of the ventilated insulation metamaterials in recent works.

Mechanisms	Metamaterials	Volume (mm^3)	Frequency range corresponding to the STL	Ventilation efficiency
Helmholtz resonance metamaterials	The slit-type Helmholtz resonator (Ref. 21)	2.16×10^6	Reach 30 dB over the target working band of 0.48–0.95 kHz,	…
	An ultra-open ventilated metamaterial absorber (Ref. 40)	8.79×10^6	The effective bandwidth is 476–726 Hz, with an average absorption efficiency of 69.3%.	62.3%–81.8%
	The double Helmholtz resonator with embedded apertures (Ref. 41)	3.49×10^6	Achieves an average 82% absorption in the range of 450 −860 Hz	51.4%–67.3%
Membrane-type acoustic metamaterials	A massless membrane acoustic metamaterial (Ref. 33)	…	The core adjustable range is 325–695 Hz, and the peak absorption coefficient is ≥95.9%.	…
	The perforated and constrained acoustic metamaterial (Ref. 34)	123	20 dB at 430 Hz	1.68%

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TABLE VI. (*Continued.*)

Mechanisms	Metamaterials	Volume (mm^3)	Frequency range corresponding to the STL	Ventilation efficiency
Membrane-coupled Helmholtz resonator metamaterials	The meta-atoms (Ref. 37)	…	The effective bandwidth is 466.5–533.5 Hz, and the frequency range with absorption coefficient ≥50%	…
	The dual-local resonance acoustic metamaterial. (Ref. 42)	0.75×10^6	10 dB within 200–240 Hz and 530–570 Hz	…
	HRDMR (this work)	0.42×10^6	It is evident that the HRDMR array achieves STL > 10 dB in the frequency range of 489–1058 Hz	95.8%–98.3%

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